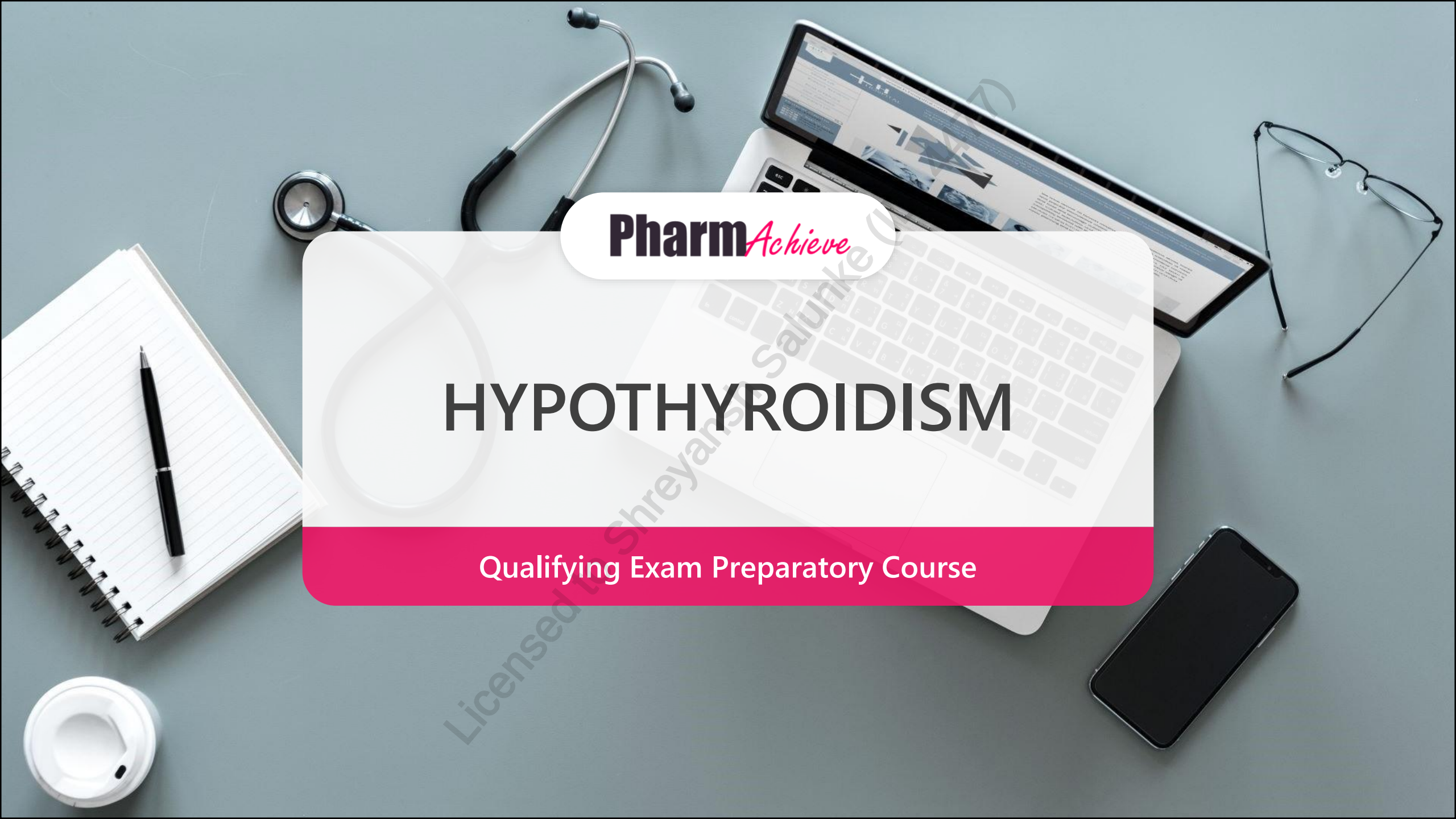


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

HYPOTHYROIDISM

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

Anti-TPO	Anti-thyroid Peroxidase
CVD	Cardiovascular Disease
ft3	Free Triiodothyronine
ft4	Free Thyroxine
PTU	Propyl-thiouracil
T3	Triiodothyronine
T4	Thyroxine
Tsat	Transferrin Saturation
TKI	Tyrosine Kinase Inhibitor
TPO	Thyroid Peroxidase
TRH	Thyrotropin Releasing Hormone
TSH	Thyroid Stimulating Hormone

Licensed to Shreyansh Salunke (ID: 4477)

PATHOPHYSIOLOGY

Primary hypothyroidism: $\uparrow$ anti-TPO, $\uparrow$ TSH, $\downarrow$ fT4

Thyroid gland disorder - Hashimoto's thyroiditis

- Autoimmune destruction of thyroid cells due to overproduction of antibodies against thyroid gland

Secondary hypothyroidism: $\downarrow \leftrightarrow$ TSH; $\downarrow$ fT4

Pituitary gland disorder (neoplasm or trauma)

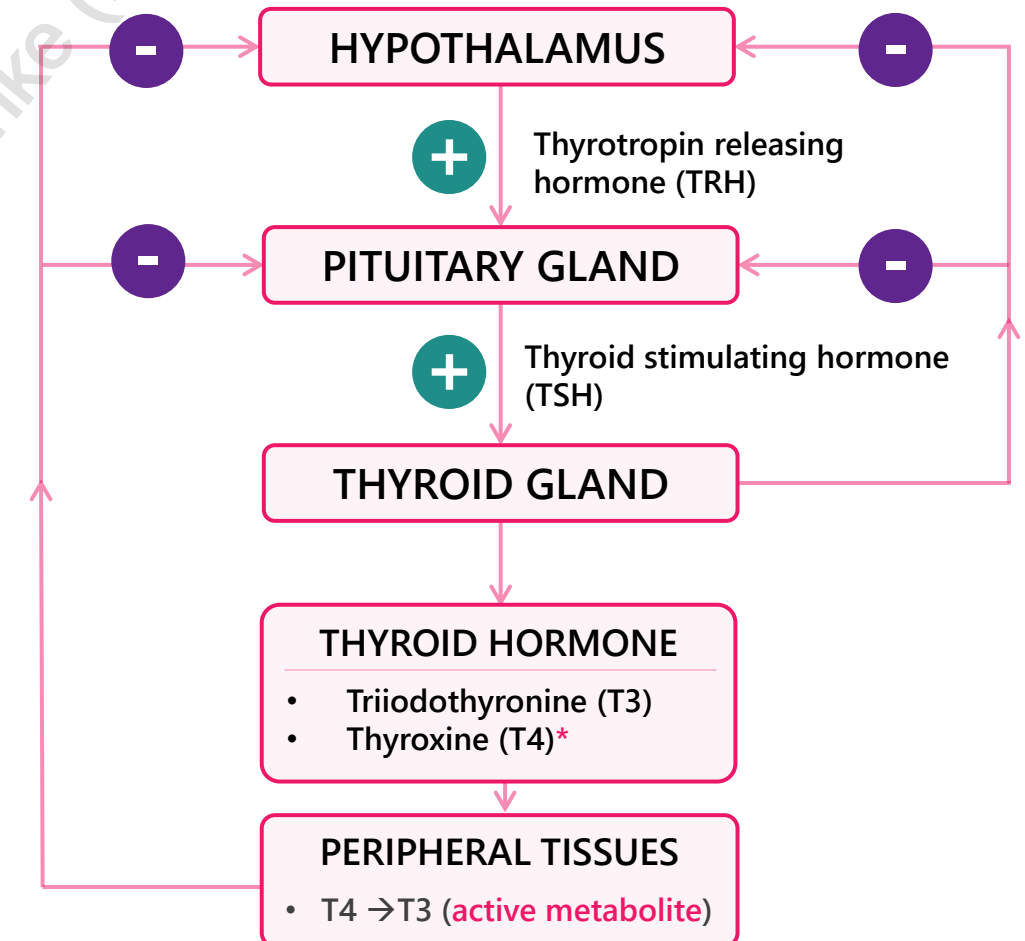
Tertiary hypothyroidism: $\downarrow \leftrightarrow$ TSH; $\downarrow$ fT4

Hypothalamus disorder (neoplasm or trauma)

Iatrogenic hypothyroidism

Due to surgery or medications

HYPOTHALAMIC-PITUITARY-THYROID AXIS



CLINICAL PRESENTATION

Slowing of metabolic activity

- Fatigue
- Cold intolerance
- Weight gain
- Constipation
- Dry skin
- Hair loss
- Slowing of mental process/memory
- Bradycardia
- Hypertension
- Macrocytic anemia
- Depression



Patients >60 years or patients with subclinical hypothyroidism may present with atypical, few, or no symptoms

DIAGNOSIS

Clinical symptoms

TSH, fT3, fT4, anti-TPO (rarely ordered)

Primary hypothyroidism: ↑TSH, ↓fT3, fT4

Subclinical hypothyroidism: ↑TSH, normal T3, T4

Consider treatment in patients with TSH >10 mU/L, abnormal lipid profile, symptoms of hypothyroidism, anti-TPO positive, or planning a pregnancy



Note: Biotin can interfere with thyroid tests. Patients taking biotin supplements should discontinue them 48 hours prior to the test.

RISK FACTORS

- Age >60 years
- Female gender
- Pregnancy
- Postpartum (<1-year post birth)
- 1st degree relative with thyroid autoimmune disease
- History of radiation or surgery (head, neck and thyroid region)
- Endocrine autoimmune conditions (e.g. type 1 diabetes, adrenal insufficiency)
- Non-endocrine autoimmune conditions (e.g. celiac disease, pernicious anemia, Sjogren's syndrome)

CAUSES

- Iodine deficiency
- Excess iodine (e.g. lithium)
- Drugs
 - Amiodarone
 - Dopamine antagonists (antipsychotics, antiemetics)
 - Sulfonylureas (gliclazide, glyburide)
 - Thionamides (methimazole, PTU)
 - Interferon alpha, monoclonal antibodies
 - Tyrosine Kinase Inhibitors (TKI)s (e.g. sunitinib)
- Iodinated contrast agents
- Thiocyanates, natural goitrogens
- Euthyroid Sick Syndrome (non-thyroidal illness)
 - Euthyroid patients with systemic disease (e.g. starvation, acute infection) that can present with abnormal lab values
 - TSH/T4 levels normalize in 4-6 weeks after treatment of underlying illness
 - Do not give exogenous thyroid hormone

GOALS OF THERAPY

- ✓ Achieve and maintain euthyroid state (normalized TSH defined in monitoring parameters)
- ✓ Manage symptoms
- ✓ Prevent complications and adverse effects from therapy



No non-drug measures! Thyroid hormone replacement is mainstay of therapy

DRUG THERAPY: SYNTHETIC T4 AND T3

Drug class, MOA	Adverse effects	Interactions	Comments
Levothyroxine (synthetic T4) - Replaces endogenous T4; converted to T3 DRUG OF CHOICE	Hyperthyroid symptoms; may worsen angina	- Absorption affected by: calcium, aluminum, magnesium, lanthanum, cholestyramine, food - INR may/may not be affected by concomitant warfarin - The following reduce serum concentrations of drug therapy: estrogen, carbamazepine, phenytoin	Administration: - Take daily, at least ½ hour before food - If not possible, take consistently with meals - Space by at least 4 hours from cholestyramine, calcium, and iron-containing products, as well as aluminum and magnesium-containing antacids - Titrate until normalized TSH and check levels in 6-8 weeks after dose changes
Liothyronine (synthetic T3) Replaces endogenous T3 (shorter-acting)			- Indicated for short-term management in patients with thyroid cancer after stopping levothyroxine - Has a short-half-life, produces large peaks in T3 in blood



Note: Avoid desiccated thyroid if possible due to inaccurate dosing

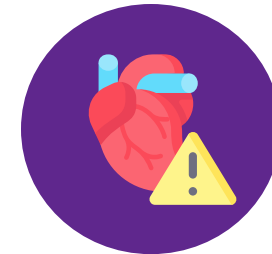
DOSING RECOMMENDATIONS: LEVOTHYROXINE

Adults:

- Start with 1.6-1.7 mcg/kg/day
- Titrate until normalized TSH and check levels in **6-8 weeks after dose changes**
- Consider dose adjustment: large body weight changes, aging, pregnancy
- **Must check thyroid panel when switching between different thyroxine brands**
- In patients with **angina or those who are elderly**, start with lower doses and monitor more frequently (every 3-6 weeks)

Monitoring:

- Initiate **cautiously** and **monitor frequently**: hypothyroidism of unknown duration, elderly, CVD
- Follow up testing:
 - 6-8 weeks after dosage adjustment
 - Every 6-12 months once stable
 - New symptoms



PREGNANCY AND BREASTFEEDING

- Thyroid binding-globulins increase during pregnancy, resulting in decreased fT4 levels due to enhanced binding of T4 by these globulins
- Requirements for thyroid replacement can increase by up to 50% to sustain TSH levels
 - Upon pregnancy confirmation, patients with hypothyroidism should **increase** their levothyroxine dose by **2 tablets per week**
- TSH levels should be obtained every 4-6 weeks following dose adjustments
- Levothyroxine is safe during pregnancy and breastfeeding
- Thyroid replacement needs return to pre-pregnancy values in the postpartum period



MYXEDEMA COMA

Definition

A severe state of hypothyroidism (medical emergency) characterized by hypothermia, extreme symptoms of hypothyroidism, and altered level of consciousness (delirium or coma)
Associated with a mortality rate of 60-70%

Causes

- Longstanding hypothyroidism

Management

- IV corticosteroids prior to administering levothyroxine
- IV levothyroxine loading dose followed by maintenance doses. Patients can be switched to oral levothyroxine once clinically improved
- Supportive therapy to address hypothermia, hypovolemia, hypoglycemia, and hypoventilation



Note: Thyroid levels should be checked more frequently, every 1-2 days

MONITORING PARAMETERS

Efficacy Endpoints	
Parameter and degree of change	Timeframe
TSH: 0.34-5.6 mIU/L *Free T4: 7.0-17.0 pmol/L *Free T3: 3.3-6.0 pmol/L	Steady state: 6-8 weeks Once stabilized, check levels q6-12 months
Improve symptoms of hypothyroidism	Can be seen in 2-3 weeks Peak effect may take 4-6 weeks

Safety Endpoints	
Parameter and degree of change	Timeframe
Prevent/minimize symptoms of thyroid over-replacement with slow titration	Duration of therapy

* Free T4 and T3 should not be used for screening or monitoring in patients with known primary hypothyroidism (unless in the presence of pituitary or hypothalamic disease)



Titrate dose every 4-6 weeks

TAKEAWAY

- **Primary hypothyroidism:** elevated TSH
- **Secondary hypothyroidism:** low or normal TSH
- **Treatment of choice: Levothyroxine**
- **Levothyroxine:** dose is 1.6-1.7 mcg/kg/day. Take at least ½ hour before food and space at least 4 hours from cholestyramine, calcium- and iron-containing products.
 - Do a blood test if switching between brands e.g. Synthroid® vs. Eltroxin®
- **Liothyronine:** used in thyroid cancer after levothyroxine, short-half-life, produces large peaks in T3 in blood. Does not interact with iron.
- Titrate levothyroxine dose every 4-6 weeks in the initial dose adjustment period
- Monitor thyroid blood tests every **6-8 weeks** until patient is stabilized
- Then monitor every **6-12 months**

CASE 1

DS is a 67-year-old female, retired housekeeper with long standing stable hypothyroidism. She was diagnosed with iron deficiency anemia 3 months ago. Prior to her diagnosis, she felt tired, lightheaded and short of breath on exertion. Her doctor advised her to start taking polysaccharide iron complex. She has been adherent with her iron supplementation and tolerates it well. However, she continues to feel tired and cold and has noticed some weight gain. She has also been feeling “brain fog” and has noticed her concentration is not as good as it used to be. She is here to see you for some herbal remedies to improve her concentration. She does not smoke or drink alcohol. She drinks 1 cup of coffee a day. She is not allergic to anything. She has hypertension, dyslipidemia, osteoporosis, diabetes, hypertrophic cardiomyopathy and anemia. Her mother had hypothyroidism (Hashimoto’s thyroiditis). She takes sitagliptin/metformin 50 mg/1000 mg BID, levothyroxine 0.088 mg daily, atorvastatin 40 mg daily, lansoprazole 30 mg daily, losartan 50 mg daily, nifedipine 60 mg daily and Prolia® (denosumab) every 6 months. She also takes calcium/vitamin D 1 tablet daily, vitamin C 500 mg daily and polysaccharide iron complex 50 mg daily.

Objective findings:

- TSH = 6.62 (0.34-5.6 mIU/L)
- fT3 = 2.8 (3.3-6.0 pmol/L)
- Ferritin = 70 (54-690 pmol/L)
- fT4 = 7.6 (7-17 pmol/L)
- Hb = 124 (120 to 160 g/L)
- tSat = 0.22 (0.20-50)

1. Further details that would be helpful to ask DS about include all the following EXCEPT:

- a) The exact dose of calcium and vitamin D
- b) The timing of her medications
- c) Whether she takes some of her medications with food
- d) Whether she takes her medications close to the time she drinks coffee



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2. Upon further questioning, DS reports that she takes her thyroid medication first thing in the morning, then waits 1 hour to eat breakfast and have tea. She then takes the rest of her medications except atorvastatin and calcium which she takes with dinner. Which of the following is true?
- a) There are no issues with DS’s medication timing
 - b) DS should space her levothyroxine from iron and calcium supplements by 4 hours
 - c) DS should take her iron supplement at dinner
 - d) DS should space her levothyroxine from food and other medications by 2-4 hours



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3. What is the next best course of action?

- a) Increase dose of levothyroxine to 0.1 mg once daily and repeat TSH, fT3, fT4 in 4 weeks
- b) Increase dose of levothyroxine and stop iron supplement
- c) Make no change to levothyroxine dose, advise DS to take iron supplement at bedtime and repeat TSH, fT3, fT4 in 6 weeks
- d) Make no change to levothyroxine dose, advise DS to take iron supplement with dinner and repeat TSH, fT3, fT4 in 6 weeks



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4. Which of the following is NOT a risk factor for DS’ hypothyroidism?

- a) Female gender
- b) Age > 65
- c) Diabetes
- d) Family history



CASE 2

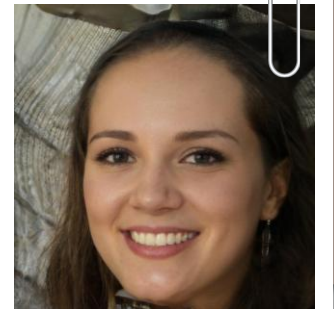
GD is a 34-year-old female, who works as a retail manager and is a mother of 2 children. She recently moved to Canada from the USA. She was previously taking levothyroxine 0.125 mg but stopped about 3 months ago due to loss of health benefits and lack of follow-up. She has been feeling tired, her "body feels off", she is sleeping more, cannot tolerate the cold weather and has noticed hair loss. She also feels that her neck is growing and noticed that her eyes are puffier. She presents to the ER today as she does not have a family doctor. Her past medical history is significant for Grave's disease, radiation at age 19, an appendectomy and 2 C-Sections. She previously had a rash after taking amoxicillin. She uses NuvaRing® (etonogestrel/ethinyl estradiol) 0.12 - 0.015 mg / 24 h and does not take anything over-the-counter. She used to use cocaine, smoke cigarettes and drink alcohol but quit 3 years ago.

Objective findings:

- BP = 118/79
- HR = 72 bpm
- No thyroid mass palpable, not enlarged
- TSH = 6.87 (0.34-5.6 mIU/L)
- fT4 = 12 (7.0-17.0 pmol/L)
- fT3 = 3.7 (3.3-6.0 pmol/L)
- Wt = 68.7 kg

1. Select the correct statement

- a) GD is suffering from secondary hypothyroidism
- b) GD is suffering from iatrogenic hypothyroidism
- c) Anti-TPO testing was needed prior to initiation of therapy with levothyroxine
- d) GD has a relapse of hyperthyroidism



CASE 2

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2. GD is started on levothyroxine 112 mcg po daily and follows up with her new family doctor 8 weeks later. She has a positive result for anti-TPO, how does this affect your treatment plan?
 - a) It requires an increase in dose of levothyroxine
 - b) It requires discontinuing levothyroxine because she has subclinical hypothyroidism
 - c) It confirms diagnosis of primary hypothyroidism and requires levothyroxine dose reduction
 - d) It does not affect it



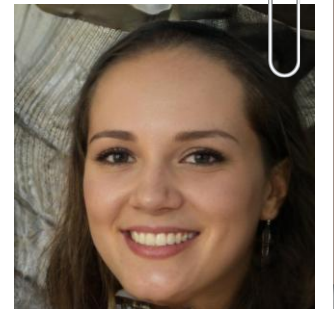
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3. GD's doctor repeats her bloodwork 8 weeks later revealing the following results: TSH = 5.37, fT4 = 15, fT3 = 5.53. All the following are appropriate next steps EXCEPT:
- a) Maintain levothyroxine dose at 112 mcg and repeat blood work in 1 week
 - b) Increase levothyroxine dose to 125 mcg and repeat blood work in 6 weeks
 - c) Review adherence of medications
 - d) Review and ensure appropriate timing and spacing of medications



CASE 2

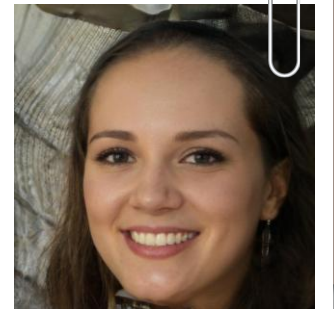
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4. What symptom would indicate that GD has been overtreated for her hypothyroidism?

- a) Hair loss
- b) Weight gain
- c) Fatigue
- d) Palpitations



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CHANGE LOG

July 2025

- Slide 10 & 15: Levothyroxine to be separated from cholestyramine by at least 4 hours

October 2025

- Slides 14, 16-23: Updated reference ranges to align with the Medical Council of Canada list of normal lab values
- Slide 24: Updated references

December 2025

- Slide 10: Added aluminum and magnesium-containing antacids to separate administration

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 14: Added note on free T4 and T3 use
- Slide 25: Updated references